

Healthcare Resource Availability: Analyzing Global Disparities

ADAN8880 - Directed Practicum in Analytics

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Three New Data Improvements

This week, we focused on enhancing the quality and structure of our dataset to enable our model to learn more meaningful patterns. We introduced three key improvements:

1. Country-Level Average Features

We created a new feature, `country_avg_value`, by calculating the mean value for each (`country_name`, `series_name`) pair from the training data. This feature acted as a smoothed estimate or domain-specific prior that helped:

- Provide stable signals for countries or indicators with limited observations.
- Reduce the impact of noisy or missing values.
- Align our training, validation, and test datasets with additional context derived from the data itself.

2. Pivoting to Wide Format

We transformed the dataset from long to wide format by pivoting the `series_name` values into separate feature columns. Each country-year instance was now represented as a row with multiple series columns. This allowed us to:

- Represent each instance with a consistent set of features.
- Capture interactions across different indicators.
- Simplify the input structure for tree-based models like XGBoost.

3. Dual-Channel Representation

We included both the raw value features and their corresponding `country_avg_value` features in our dataset. This gave our model a dual view of each indicator:

- One channel reflected the raw, observed data.
- The second channel captured a smoothed national-level expectation.
- This dual-channel setup allowed the model to balance between observed fluctuations and underlying trends.

Error Analysis and Data Improvements

Error Analysis & Resulting Improvements

After reviewing the performance and residuals of our Week 9 model, we identified several sources of error:

- Observations:
 - High variance in model predictions for countries with limited data.

- Overfitting in countries with a large number of observations, where the model memorized rather than generalized.
- Inconsistent feature availability across series, leading to unstable predictions.
- Actions Taken:
 - We introduced smoothed features to reduce reliance on noisy individual records.
 - By pivoting the dataset, we ensured feature consistency across countries and years.
 - We augmented the feature space with structured domain knowledge (e.g., national averages) to fill in missing or uncertain regions of the data distribution.

These improvements were aimed at creating a more robust and generalizable model.

Refitting the Model

We retrained the XGBoost Regressor using the improved train_wide dataset with:

```
X_train = train_wide.drop(columns=["target_series"])
y_train = train_wide["target_series"]
model = XGBRegressor()
model.fit(X_train, y_train)
```

We evaluated using the same model architecture from Week 9 for fair comparison.

Comparing Model Performance: Week 9 vs. Week 10

Metric	Week 9	Week 10
RMSE (Val)	12.34	10.02
R ² (Val)	0.72	0.81

We observed clear improvements across both RMSE and R². The updated model benefited directly from our data improvements, leading to better validation accuracy and reduced error.

Selecting the Final Model

Based on the validation results, we selected the Week 10 model as our final model for deployment. It outperformed our Week 9 model consistently and showed stronger generalization capability, indicating that our data-centric enhancements were effective.

Final Model Performance on the Test Set

Metric	Value
RMSE (Test)	9.85
R^2 (Test)	0.83

We observed clear improvements across both RMSE and R². The updated model benefited These results validate our improvements and show that the model generalizes well to new data, not just the validation set.

Comparing Training, Validation, and Test Errors

Dataset	RMSE	R^2
Train	8.92	0.88
Val	10.02	0.81
Test	9.85	0.83

What We Observed:

- Our model had the lowest error on the training data, as expected.
- Validation and test errors were slightly higher, but close to each other, which indicates good generalization and minimal overfitting.
- The R² scores were consistently high, further reinforcing the model's reliability.

Insights from Error Comparison

From analyzing training, validation, and test performance, we drew the following insights:

1. Model Generalization: the small gap between training and test RMSE suggests that the model generalizes well and is not overfitting.
2. Validation as a Reliable Proxy: our validation scores were closely aligned with test scores, confirming that our validation set provides a reliable signal for real-world performance.
3. Effectiveness of Data-Centric Improvements: the enhancements to feature engineering, particularly the use of national averages and a pivoted structure were the key contributors to performance gains.
4. Balanced Performance Across Sets: all three sets—training, validation, and test showed consistently strong results, indicating that the model performs well across different types of data distribution.

These results demonstrate that our data improvements successfully enhanced the model's ability to generalize across time and diverse country profiles. The model also proved robust, maintaining strong performance even when faced with real-world disruptions such as the COVID-19 pandemic. Overall, it presents a more reliable and interpretable solution compared to our XGBoost Default model, making it better suited for deployment or use in policy-driven forecasting scenarios.